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$$f(x) = x^x \quad (x > 0)$$

$$\ln f(x) = \ln(x^x) = x \ln x$$

$$f(x) = e^{x \ln x}$$

$$f'(x) = e^{x \ln x} (x \ln x)'$$

$$(x \ln x)' = 1 \cdot \ln x + x \cdot \frac{1}{x}$$

$$= \ln x + 1$$

$$f'(x) = e^{x \ln x} (\ln x + 1)$$

$$= x^x (\ln x + 1)$$

$$f'(x) = 0$$

$$x^x (\ln x + 1) = 0$$

$$\ln x + 1 = 0$$

$$\ln x = -1$$

$$x = e^{-1}$$

| $x$         | 0 | $e^{-1}$   | $+\infty$  |
|-------------|---|------------|------------|
| $\ln x + 1$ | - | 0          | +          |
| $f'(x)$     | - | 0          | +          |
| $f(x)$      |   | $\searrow$ | $\nearrow$ |

$$f(e^{-1}) = (e^{-1})^{e^{-1}} = e^{-1/e}$$

|                              |
|------------------------------|
| minimum $(e^{-1}, e^{-1/e})$ |
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